A RIGOROUS LOWER BOUND FOR SITE PERCOLATION ON THE SIMPLE CUBIC LATTICE: $p_c^{\mathrm{site}}(\mathbb{Z}^3) \geq 0.263376$

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1. Summary

Theorem 1.1. *The critical probability of site percolation on $\mathbb{Z}^3$ satisfies*

$$p_c^{\mathrm{site}}(\mathbb{Z}^3) \geq 263376/10^6 = 0.263376.$$

Context. Georgakopoulos and Panagiotis proved $p_c^{\mathrm{site}}(\mathbb{Z}^3) > 0.2522$ [2, Theorem 7.1], the best rigorous lower bound we know of; their argument gives $0.25226\ldots$ (Remark 7.2). Theorem 1.1 improves it by about 0.011. Numerical estimates place $p_c^{\mathrm{site}}(\mathbb{Z}^3)$ near 0.3116.

Idea. In the open cluster of the origin, give every site at graph distance $k \geq 1$ from 0 a parent: its neighbour at distance $k-1$ with the smallest direction index in the fixed order $+x, -x, +y, -y, +z, -z$. This gives a spanning tree, and every path in it from the origin is a neighbour-avoiding walk. Along such a path some sites are forced to be closed:

- every site adjacent to two path sites at least three steps apart (it would be a shortcut);
- at a turn whose first step lies on an earlier axis than its second step (in the order x, y, z), the corner site, which would otherwise have been the preferred parent.

If the cluster is infinite, the tree has an infinite ray. Its first N steps form such a path, and the rest of the ray is an infinite open path that avoids the path and its forced-closed sites. The two events involve disjoint sets of sites, which gives $\theta(p) \leq K_N(p)\,\theta(p)$ with an explicit polynomial $K_N(p) = \sum_f c_f\, p^{N+1}(1-p)^f$. If $K_N(p) < 1$, then $\theta(p) = 0$.

Certificate. The integers c_f for $N = 20$ come from an exhaustive enumeration covering all 3,516,072,837,894 neighbour-avoiding walks of length 20, through one representative per sign-flip orbit. They are listed in Table 1, so anyone can check the final inequality with exact arithmetic:

$$K_{20}(263376/10^6) = 0.999971526023\ldots < 1.$$

The same table gives $K_{20}(5/19) < 1$ (Remark 7.1).

How to check. Code, tables and checkers are available at `https://kaust-a-laut.github.io` (folder `results/z3-site-lower-bound-0.263376`). The same package is also available as a draft pull request, `https://github.com/elder-plinius/V3R1T4S/pull/571`. Its command `sh verify-it-yourself/run_checks.sh` takes about half a minute and needs only Python's standard library. It checks:

- the enumerator against a direct brute-force implementation of Definition 3.1, for $3 \leq N \leq 8$;
- every table against the published counts of neighbour-avoiding walks (OEIS A174319);
- the certificate, in exact integers.

With `FULL=1`, it also rebuilds the enumerator from source and regenerates the $N = 20$ table byte for byte, in about 2.5 hours on 6 cores.

Status. This work was found and checked with substantial assistance from an AI system (Claude, by Anthropic). Three separate AI reviews checked the argument, the literature and

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the computation. During one of them a second enumerator, with a different symmetry reduction, was written; it reproduced the whole $N = 20$ table exactly. A further adversarial AI review of the written proof found no error affecting the theorem. It found one false remark, harmless for the walk lengths used, and several gaps in the exposition; all are corrected here (Section 8). No human expert has checked the proof or the computation yet.

2. Setting

Consider site percolation on the nearest-neighbour graph $\mathbb{Z}^3$ with parameter $p \in [0,1]$. Under $\mathbb{P}_p$, sites are open independently with probability p. Let $C(0)$ be the open cluster of 0, which is empty if 0 is closed. Let $\theta(p) = \mathbb{P}_p(|C(0)| = \infty)$ and $p_c^{\text{site}}(\mathbb{Z}^3) = \sup\{p : \theta(p) = 0\}$. Write $x \sim y$ when x and y are neighbours. For $x \in C(0)$, let $d(x)$ be the graph distance from 0 to x in the subgraph induced by the open sites; d is defined only on $C(0)$.

The six unit vectors are $e_0, \dots, e_5 = +x, -x, +y, -y, +z, -z$. Put $\iota(e_i) = i$, and let the axis of e_{2a} and e_{2a+1} be $\text{ax}(e_{2a}) = \text{ax}(e_{2a+1}) = a$, for $a \in \{0,1,2\}$. Every direction index of axis a is smaller than every direction index of axis b when $a < b$.

A *walk* is a sequence $\gamma = (v_0, \dots, v_N)$ with $v_{k-1} \sim v_k$. Its *steps* are $s_k = v_k - v_{k-1}$ for $1 \le k \le N$.

Definition 2.1. A walk $\gamma = (v_0, \dots, v_N)$ is a *neighbour-avoiding walk* (NAW) if it is self-avoiding (the v_i are distinct) and $v_i \sim v_j$ only when $|i-j| = 1$. Let a_N be the number of N-step NAWs from 0.

For $N \ge 3$, the adjacency condition already implies self-avoidance. For $N = 2$ it does not: $(0, e_0, 0)$ satisfies it. The sequence (a_N) is OEIS A174319 [3].

3. The parent tree and the forced-closed sites

Let $x \in C(0) \setminus \{0\}$ and $k = d(x) \ge 1$. The set $P(x) = \{w \sim x : w \in C(0),\ d(w) = k-1\}$ is nonempty: it contains the second-to-last site of a shortest open path from 0 to x. Define $\text{parent}(x)$ as the element w of $P(x)$ with the smallest $\iota(w - x)$. It is unique, because distinct neighbours of x lie in distinct directions.

Since $d(\text{parent}(x)) = d(x) - 1$, following parents from x reaches 0 after $d(x)$ steps. So the edges $\{x, \text{parent}(x)\}$ form a tree T with vertex set $C(0)$ and root 0. Every vertex of T has degree at most 6, so T is locally finite.

A *tree path* is a sequence $(v_0, \dots, v_N)$ with $v_0 = 0$, $v_k \in C(0)$ for $0 \le k \le N$, and $\text{parent}(v_k) = v_{k-1}$ for $1 \le k \le N$. In particular, a tree path exists only when 0 is open.

Definition 3.1. Let $\gamma = (v_0, \dots, v_N)$ be an N-step NAW from 0. For $1 \le k \le N-1$, put $u_k = v_{k-1} + v_{k+1} - v_k$. Define

$$S(\gamma) = \{w \notin \gamma : w \sim v_i \text{ and } w \sim v_j \text{ for some } 0 \le i < j \le N \text{ with } j - i \ge 3\} \quad \text{(shortcut sites)},$$

$$Q(\gamma) = \{u_k : 1 \le k \le N-1,\ s_{k+1} \ne s_k,\ \iota(u_k - v_{k+1}) < \iota(v_k - v_{k+1})\} \quad \text{(turn corners)},$$

$F(\gamma) = S(\gamma) \cup Q(\gamma)$, and

$$E(\gamma) = \{y \sim v_N : y \notin \gamma,\ y \not\sim v_i \text{ for all } i < N,\ y \notin F(\gamma)\}.$$

Thus $E(\gamma)$ is the set of sites $y \notin F(\gamma)$ for which $(v_0, \dots, v_N, y)$ is an $(N+1)$-step NAW.

Lemma 3.2 (Turns)**.** *Let γ be a NAW and $1 \le k \le N-1$ with $s_{k+1} \ne s_k$. Then:*

- *s_{k+1} is perpendicular to s_k;*
- *$u_k \sim v_{k-1}$, $u_k \sim v_{k+1}$ and $u_k \notin \gamma$;*
- *the condition $\iota(u_k - v_{k+1}) < \iota(v_k - v_{k+1})$ holds if and only if $\text{ax}(s_k) < \text{ax}(s_{k+1})$.*

In particular, $F(\gamma) \cap \gamma = \emptyset$.

Proof. Since γ is self-avoiding, $v_{k+1} \neq v_{k-1}$, so $s_{k+1} \neq -s_k$. Together with $s_{k+1} \neq s_k$, this makes the two unit vectors perpendicular.

Next, $u_k - v_{k-1} = s_{k+1}$ and $u_k - v_{k+1} = -s_k$ are unit vectors, so $u_k \sim v_{k-1}$ and $u_k \sim v_{k+1}$. Suppose $u_k = v_m$. Then the NAW property gives $|m - (k-1)| = 1$ and $|m - (k+1)| = 1$, so $m = k$. But $u_k - v_k = s_{k+1} - s_k \neq 0$, a contradiction.

Finally, $u_k - v_{k+1} = -s_k$ lies on the axis of s_k, and $v_k - v_{k+1} = -s_{k+1}$ lies on the axis of s_{k+1}. These axes differ, so the comparison of direction indices is decided by the axes alone.

For the last claim: $S(\gamma)$ excludes γ by definition, and $Q(\gamma)$ does by the second item. □

Lemma 3.3 (Tree paths)**.** *Let $\gamma = (v_0, \dots, v_N)$ be a tree path. Then:*

(a) *every v_k is open and $d(v_k) = k$; in particular, the v_k are distinct;*
(b) *γ is a NAW;*
(c) *every site of $F(\gamma)$ is closed.*

Proof. (a) Each v_k is in $C(0)$, so it is open, and $d(v_k) = d(\text{parent}(v_k)) + 1 = d(v_{k-1}) + 1$, with $d(v_0) = 0$.

(b) Self-avoidance follows from (a). If $v_i \sim v_j$ with $j \geq i + 2$, then both are open, so $d(v_j) \leq d(v_i) + 1 = i + 1 < j$, contradicting (a).

(c) Shortcut sites: let $w \in S(\gamma)$, with $w \sim v_i$, $w \sim v_j$ and $j - i \geq 3$. If w were open, then $w \in C(0)$ and $d(w) \leq i + 1$, so $d(v_j) \leq i + 2 < j$, contradicting (a).

Turn corners: let $u = u_k \in Q(\gamma)$. By Lemma 3.2, $u \sim v_{k-1}$ and $u \sim v_{k+1}$. If u were open, then $u \in C(0)$ and $d(u) \leq d(v_{k-1}) + 1 = k$. Also $k + 1 = d(v_{k+1}) \leq d(u) + 1$, so $d(u) = k$ and $u \in P(v_{k+1})$. Since $v_k \in P(v_{k+1})$ as well, and $\iota(u - v_{k+1}) < \iota(v_k - v_{k+1})$, we get $\text{parent}(v_{k+1}) \neq v_k$, contradicting the definition of a tree path. □

4. The finite criterion

Let γ be an N-step NAW from 0. Put

$$A_\gamma = \{\text{every site of } \gamma \text{ is open}\} \cap \{\text{every site of } F(\gamma) \text{ is closed}\}.$$

By self-avoidance and Lemma 3.2, this event concerns $N + 1 + |F(\gamma)|$ distinct sites, so

$$\mathbb{P}_p(A_\gamma) = p^{N+1}(1-p)^{|F(\gamma)|}, \quad \text{with } 0^0 = 1.$$

Let G_γ be the subgraph of $\mathbb{Z}^3$ induced by $\mathbb{Z}^3 \setminus (\gamma \cup F(\gamma))$. For $y \in E(\gamma)$, which is a vertex of G_γ, define

$$B_{\gamma,y} = \{\text{the open cluster of } y \text{ in } G_\gamma \text{ is infinite}\},$$

which is empty if y is closed.

Lemma 4.1 (Independence)**.** *$B_{\gamma,y}$ is determined by the states of the sites outside $\gamma \cup F(\gamma)$. Hence it is independent of A_γ under $\mathbb{P}_p$, and $\mathbb{P}_p(B_{\gamma,y}) \leq \theta(p)$.*

Proof. For $n \geq 1$, let $\Lambda_n = \{x : \|x - y\|_\infty \leq n\}$, and let B^n be the event that some path of open sites of G_γ inside Λ_n joins y to a site with $\|x - y\|_\infty = n$. Then B^n depends only on the finitely many sites of $\Lambda_n \setminus (\gamma \cup F(\gamma))$. Moreover $B_{\gamma,y} = \bigcap_n B^n$:

- if the open cluster of y in G_γ is infinite, it contains sites outside Λ_{n-1}, and a path to such a site meets $\{\|x - y\|_\infty = n\}$ for the first time inside Λ_n;
- if it is finite, it lies inside some Λ_{n-1}, and then B^n fails.

This proves the first claim, and independence follows because $\mathbb{P}_p$ is a product measure.

Finally, $B_{\gamma,y} \subseteq \{|C(y)| = \infty\}$, where $C(y)$ is the open cluster of y in $\mathbb{Z}^3$, and $\mathbb{P}_p(|C(y)| = \infty) = \theta(p)$ by translation invariance. □

Lemma 4.2 (Coverage)**.** *Let $N \geq 1$. If $|C(0)| = \infty$, then there are a tree path γ of length N and a site $y \in E(\gamma)$ such that A_γ and $B_{\gamma,y}$ both hold.*

Proof. T is an infinite, locally finite tree, so by Kőnig's lemma it contains an infinite ray $0 = v_0, v_1, v_2, \ldots$ with $\text{parent}(v_{m+1}) = v_m$ for all m. Put $\gamma = (v_0, \ldots, v_N)$ and $y = v_{N+1}$. Then γ is a tree path, so A_γ holds by Lemma 3.3(a) and (c).

Since $(v_0, \ldots, v_{N+1})$ is also a tree path, Lemma 3.3 gives the following:

- it is a NAW, so $y \not\sim v_i$ for $i < N$;
- $d(y) = N + 1 > N \geq d(v_i)$ for every $i \leq N$, so $y \notin \gamma$;
- y is open, while every site of $F(\gamma)$ is closed, so $y \notin F(\gamma)$.

Hence $y \in E(\gamma)$.

The sites v_m, $m \geq N+1$, are distinct and open. They are not in γ, because $d(v_m) = m > N$. They are not in $F(\gamma)$, because they are open. So they are vertices of G_γ, consecutive ones are adjacent, and the open cluster of y in G_γ is infinite. Thus $B_{\gamma,y}$ holds. □

Lemma 4.3. *For every $N \geq 1$ and $p \in [0,1]$,*

$$\theta(p) \leq K_N(p)\,\theta(p), \qquad K_N(p) = \sum_\gamma |E(\gamma)|\, p^{N+1}(1-p)^{|F(\gamma)|},$$

where the sum is over the N-step NAWs γ from 0.

Proof. By Lemma 4.2 and Lemma 3.3(b), $\{|C(0)| = \infty\}$ is contained in the finite union of the events $A_\gamma \cap B_{\gamma,y}$, over NAWs γ and $y \in E(\gamma)$. By Lemma 4.1, $\mathbb{P}_p(A_\gamma \cap B_{\gamma,y}) = \mathbb{P}_p(A_\gamma)\mathbb{P}_p(B_{\gamma,y}) \leq p^{N+1}(1-p)^{|F(\gamma)|}\theta(p)$. Now sum over the pairs (γ, y). □

Corollary 4.4. *If $K_N(p) < 1$ for some $N \geq 1$, then $\theta(p) = 0$, and hence $p_c^{\text{site}}(\mathbb{Z}^3) \geq p$.*

Proof. $\theta(p)$ is finite, so $(1 - K_N(p))\theta(p) \leq 0$ forces $\theta(p) = 0$. Then $p \in \{p' : \theta(p') = 0\}$, so $p \leq p_c^{\text{site}}(\mathbb{Z}^3)$. Monotonicity of θ is not needed. □

Remark 4.5. The condition $y \notin F(\gamma)$ in the definition of $E(\gamma)$ is implied by the other conditions. Every site of $F(\gamma)$ is adjacent to some v_i with $i < N$:

- a shortcut site is adjacent to some v_i with $i \leq N - 3$;
- a corner u_k is adjacent to v_{k-1}, and $k - 1 \leq N - 2$.

Hence $|E(\gamma)|$ is the number of one-step NAW extensions of γ, and $\sum_\gamma |E(\gamma)| = a_{N+1}$. This gives an external check of the computation against OEIS A174319. Nothing in this section is specific to $N = 20$.

5. Computing K_N

For $f \geq 0$, let c_f be the sum of $|E(\gamma)|$ over the N-step NAWs γ from 0 with $|F(\gamma)| = f$. Then $K_N(p) = \sum_f c_f\, p^{N+1}(1-p)^f$.

Lemma 5.1 (Sign flips)**.** *Let $\mathcal{G} \cong (\mathbb{Z}_2)^3$ act on $\mathbb{Z}^3$ by changing the signs of coordinates. For $g \in \mathcal{G}$ and an N-step NAW γ from 0, the walk $g\gamma$ is an N-step NAW from 0, with $F(g\gamma) = gF(\gamma)$ and $E(g\gamma) = gE(\gamma)$.*

Proof. g is linear and is a graph automorphism of $\mathbb{Z}^3$ that fixes 0. So it preserves self-avoidance, adjacency, the set S, and the adjacency conditions defining E.

It maps each step s_k to gs_k, so it preserves which steps are equal and the axis of every step. By linearity, $u_k(g\gamma) = g\,u_k(\gamma)$. By Lemma 3.2, the corner condition depends only on the axes of s_k and s_{k+1}, which g preserves. □

Let $A(\gamma)$ be the set of axes used by the steps of γ. A flip supported on unused axes fixes every vertex of γ. A flip that changes the sign of a used axis a changes the first step along a. So the stabiliser of γ is the subgroup of flips supported on the unused axes, and the orbit of γ has $2^{|A(\gamma)|}$ elements. Exactly one of them, the *canonical* one, has a positive first step along every used axis. Therefore

$$c_f = \sum_{\gamma \text{ canonical},\ |F(\gamma)|=f} 2^{|A(\gamma)|}\,|E(\gamma)|.$$

The enumerator (`object/code/par/main.go` in the package) enumerates the canonical N-step NAWs by depth-first search, taking a step along a not-yet-used axis only in the positive direction. It maintains a set of flagged sites:

- when v_j is appended, every off-path neighbour z of v_j that has a path neighbour v_i with $i \le j-3$ is flagged (a shortcut site);
- if the walk turns at v_{j-1} and the condition of Definition 3.1 holds, the corner u_{j-1} is flagged.

At length N, it counts $|E|$, the neighbours of v_N that are off the path, have no path neighbour other than v_N, and are not flagged, and it adds $2^{|A|}|E|$ to $c_{|F|}$.

A site is flagged at most once, so $|F|$ counts distinct sites, and flags are removed again when the search backtracks. The flagged set equals $F(\gamma)$ of the final walk:

- every site of $F(\gamma)$ is flagged: a shortcut site with witnesses v_i, v_j ($j-i \ge 3$) when v_j is appended, and a corner u_k when v_{k+1} is appended;
- no flagged site is on the path when it is flagged (by definition, and by Lemma 3.2);
- a flagged site z has path neighbours v_a and v_b with $b-a \ge 2$, where appending v_b flagged it. If it later became v_m, with $m > b$, then $v_m \sim v_a$ and $m - a \ge 3$, which is impossible for a NAW. So z stays off the path and satisfies the definition of $S(\gamma)$ or $Q(\gamma)$ for the final walk.

The enumerator is checked independently:

- the package's checker recomputes every c_f for $3 \le N \le 8$ directly from Definition 3.1, without the orbit reduction;
- all tables satisfy $\sum_f c_f = a_{N+1}$ (Remark 4.5) and reproduce the walk counts of OEIS A174319;
- a second enumerator, written separately during one of the AI reviews with a different symmetry reduction, reproduced the whole $N = 20$ table exactly.

6. The certificate

Table 1 lists the coefficients of $K_{20}(p) = \sum_f c_f\, p^{21}(1-p)^f$. The enumerator visits only canonical walks; weighted by their orbit sizes $2^{|A(\gamma)|}$, they number $a_{20} = 3{,}516{,}072{,}837{,}894$. Also $\sum_f c_f = 14{,}334{,}995{,}983{,}614 = a_{21}$. Both values agree with OEIS A174319. In the package, the table is the file `object/outputs/p20.txt`, with SHA-256 `04bce75fa19d4609b41f0ba82d0ef32d3c78259dde5f9fcd492917afdaccc989`.

What is checked by hand, and what by computer. Sections 2–5 prove Corollary 4.4 and can be checked by hand. The only computer-dependent input is Table 1: the values $c_0, \dots, c_{23}$, and the fact that $c_f = 0$ for $f > 23$. The second follows from the first: by Remark 4.5, $\sum_f c_f = a_{21}$, and the printed values already sum to a_{21}, so the remaining coefficients, which are nonnegative, vanish. The evidence for the printed values is:

- two separately written enumerators, with different symmetry reductions, produce the same table;

- a direct brute-force implementation of Definition 3.1 agrees with the enumerator for $3 \le N \le 8$;
- for every N computed, the orbit-weighted walk count and $\sum_f c_f$ agree with a_N and a_{N+1} from OEIS A174319.

The last item constrains only totals; the first two test the individual coefficients.

f	c_f	f	c_f
0	7,854	12	372,152,376,924
1	6,681,940	13	155,487,602,400
2	708,108,008	14	57,330,294,000
3	19,477,666,848	15	18,453,225,112
4	198,536,874,684	16	5,074,284,896
5	913,320,282,824	17	1,156,374,344
6	2,164,071,405,764	18	212,293,320
7	3,016,560,356,860	19	30,921,448
8	2,927,249,415,692	20	3,607,216
9	2,259,476,003,180	21	344,768
10	1,441,374,348,408	22	27,824
11	784,313,477,660	23	1,640

TABLE 1. The coefficients c_f of K_{20} (package file `object/outputs/p20.txt`).

Proof of Theorem 1.1. With the coefficients of Table 1 and $p = 263376/10^6$, exact rational arithmetic gives

$$K_{20}(263376/10^6) = 0.999971526023\ldots < 1,$$

so $1 - K_{20}(263376/10^6) \approx 2.85 \times 10^{-5}$. Corollary 4.4 then gives $p_c^{\mathrm{site}}(\mathbb{Z}^3) \ge 263376/10^6$. □

Remark 6.1. At the next grid value, $K_{20}(263377/10^6) = 1.000041779029\ldots > 1$. Each term $p^{21}(1-p)^f$ is increasing on $[0, 21/(21+f)]$, and $f \le 23$, so K_{20} is increasing on $[0, 21/44]$. Hence $263376/10^6$ is the largest multiple of 10^{-6} in $[0, 21/44]$ at which $K_{20} < 1$. This is not needed for the theorem. Since K_{20} is continuous, $K_{20} < 1$ also holds slightly above $263376/10^6$, so in fact $p_c^{\mathrm{site}}(\mathbb{Z}^3) > 263376/10^6$.

7. Remarks

Remark 7.1. $K_{20}(5/19) = 0.984758991852\ldots < 1$, so $p_c^{\mathrm{site}}(\mathbb{Z}^3) \ge 5/19 \approx 0.263158$, slightly weaker than Theorem 1.1. The value 5/19 is what the "putative lower bound" S_3/S_4 of Torquato and Jiao [4, Eq. (43) and Table III] gives for site percolation on $\mathbb{Z}^3$. They write that a rigorous proof is beyond the scope of their paper and reserved for future work, and we have not found one. Their rigorous bound [4, Eq. (29)] gives 1/6.

Remark 7.2. Georgakopoulos and Panagiotis [2] relate the exponential growth rate $\dot{a}(\mathbb{Z}^d)$ of site lattice animals to the threshold: $\dot{a}(\mathbb{Z}^d) \ge f(r(p_c^{\mathrm{site}}(\mathbb{Z}^d)))$, where $f(r) = (1+r)^{1+r}/r^r$ and $r(p) = (1-p)/p$. Combined with the bound $\dot{a}(\mathbb{Z}^3) < 9.3835$ of Barequet and Shalah [1], this gives $p_c^{\mathrm{site}}(\mathbb{Z}^3) > 0.25226535\ldots > 0.2522$. Since $f \circ r$ is decreasing, reaching 0.263376 by this route would need $\dot{a}(\mathbb{Z}^3) \le f(r(0.263376)) = 8.92724\ldots$.

Remark 7.3. We expect larger N to give better bounds, but have not proved that the bound improves with N. Ratio estimates from the tables for $N \le 20$ suggest that, with this forced structure, it levels off near 0.278. That is an estimate, not a result.

8. Relation to the package, and corrections

The package contains the proof (`object/PROOF.md`) exactly as the AI review panel checked it. This note restates that proof with the following corrections, all from a later adversarial AI review. None of them changes a number.

(1) The definition of a NAW now requires self-avoidance. `PROOF.md` defined a NAW only by "$v_i \sim v_j$ only for $|i-j|=1$" and remarked that it is then self-avoiding. That remark is false for $N=2$: the walk $(0,e,0)$ satisfies the condition. Nothing changes numerically:
   - for $N \geq 3$ the two definitions agree;
   - the six extra 2-step walks have $E=\emptyset$;
   - the enumerator and the checker both require self-avoidance;
   - the tables start at $N=3$.

(2) It is now explicit that tree paths require 0 to be open, and that d is defined only on $C(0)$.

(3) Lemma 4.2 derives $y \notin \gamma$ from $d(y) = N+1$, instead of from the faulty remark.

(4) Lemma 4.1 proves that $B_{\gamma,y}$ depends only on the sites outside $\gamma \cup F(\gamma)$, as an intersection of finite cylinder events.

(5) The corner condition is restated in terms of axes (Lemma 3.2). This makes the sign-flip invariance (Lemma 5.1) explicit, including that the corner map commutes with flips.

(6) Smaller points:
   - monotonicity of θ is not used;
   - the convention $0^0 = 1$ is stated;
   - steps are numbered $s_1, \ldots, s_N$;
   - the set of turn corners, $C(\gamma)$ in `PROOF.md`, is written $Q(\gamma)$ here, to avoid a clash with the cluster $C(0)$;
   - the certificate table is included (Table 1).

The package's `ERRATA.md` also lists package-level corrections that do not concern the proof: the citation of the previous record, a toolchain note, and line endings.

References

[1] G. Barequet and M. Shalah, *Improved upper bounds on the growth constants of polyominoes and polycubes*, arXiv:1906.11447.

[2] A. Georgakopoulos and C. Panagiotis, *On the exponential growth rates of lattice animals and interfaces*, Combin. Probab. Comput. **32** (2023), no. 6, 912–955, doi:10.1017/S0963548323000214 (arXiv:1908.03426 and arXiv:2107.06272).

[3] OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, sequence A174319, `https://oeis.org/A174319`.

[4] S. Torquato and Y. Jiao, *Effect of dimensionality on the percolation thresholds of various d-dimensional lattices*, Phys. Rev. E **87** (2013), 032149 (arXiv:1302.0332).

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